

Echoes of the Creator: An Immersive VR System for Spatial Storytelling and Empathy Towards Co-Creation

Yifan Chen
Trinity College Dublin
Dublin, Ireland
cheny41@tcd.ie

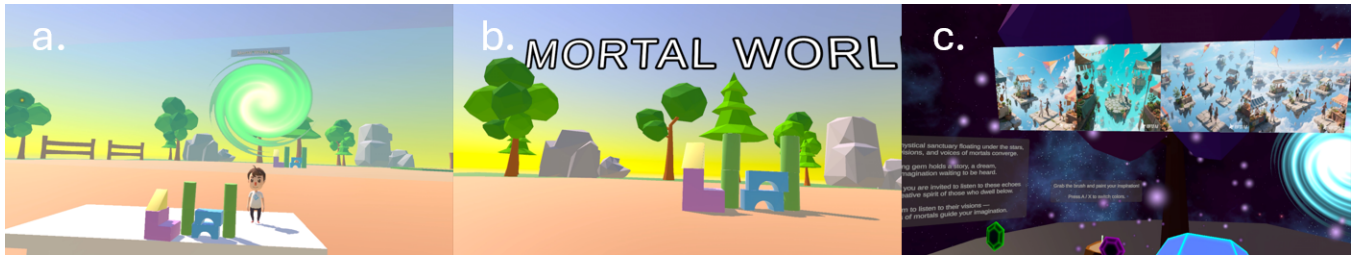


Figure 1: Overview of the three VR scenes: a. Heaven; b. Mortal World; c. Echoes Origin.

Abstract

“Echoes of the Creator” is a single-user immersive VR experience that simulates creator-user role shifts through spatial storytelling and multimodal interaction. By fostering embodied empathy and reflecting on power dynamics, it offers insights into co-creative processes. AI-driven companions guide the user and provide narrative feedback throughout the journey. The prototype lays the foundation for future multi-user experiences, inviting new forms of collaborative world-building and creative reflection.

CCS Concepts

• **Human-centered computing** → **Collaborative and social computing systems and tools**; *Interaction techniques*; • **Computing methodologies** → **Virtual reality**.

Keywords

Immersive Storytelling, Virtual Reality, Co-creation, Embodied Empathy, Role-shifting, Multimodal Interaction, AI Agent

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1 Introduction

Co-creation is increasingly valued across domains such as education, healthcare, and digital art. However, in practice, it often

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occurs under power and information asymmetries. Creators with relevant skills, such as designers and developers, retain control over the structure and tools, while the real users serve merely as passive providers of information and feedback, rather than as active co-creators. Mutual understanding becomes difficult when each party remains locked in its own experiential frame. What is missing is embodied perspective-taking, a means to inhabit the roles, responsibilities, and limitations of the other.

Virtual Reality (VR) was described as an “ultimate empathy machine” [3], providing users with a unique opportunity to experience different viewpoints from a first-person perspective in an immersive virtual environment [4]. When coupled with immersive storytelling and interactive world-building, VR can become a space where role, agency, and affective experience are fluid and reconfigurable.

To explore this possibility, “Echoes of the Creator”, a conceptual prototype, was proposed to simulate dynamic role shifts between a god-like creator and a mortal inhabitant through spatial storytelling. By enabling the user to alternate between constructing a world from above and inhabiting it from within, the system employs body-size transformation and spatial narrative design as metaphors for power, responsibility, and lived experience. These embodied shifts are intended not only to understand the feelings of the creator and user but also to experience those feelings in a physical way.

The transformation of form is the flow of player identity, responsibility and perspective. By staging this embodied transformation, the project invites us to consider the following questions:

- Can creators truly experience the spaces they design from the user’s perspective?
- Can stakeholders better understand each other’s design constraints through embodied role-play?
- Will creators be more willing or able to act on feedback after such perspective-taking?

2 Narrative Architecture

The core of “Echoes of the Creator” lies in the dual-role design:

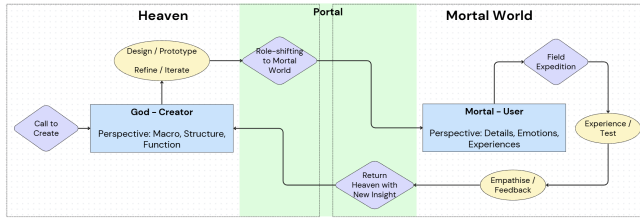


Figure 2: Narrative architecture combining Hero's Journey with a Design Thinking loop.

- **God:** a divine creator assembling building blocks for mortals, holding a macro perspective. From this scale, mortal habitats appear as manipulable toys.
- **Mortal:** a human inhabitant who explores these constructions from within, focusing more on details, emotions and experiences.

The narrative architecture (see Fig. 2) draws from the Hero's Journey [1], embodied and experiential narratives, and cyclical narrative structures. The Hero's Journey forms the backbone: *Call to Create* → *Role-shifting to Mortal World* → *Field Expedition* → *Return to Heaven with New Insight*. Role-shifting serves as a storytelling technique to drive the story through changes in physicality. This physical engagement makes participants more aware of the differences in experience between different identities, thereby enhancing their understanding of the design challenges and experience limitations faced by each other's characters.

Rather than following a linear plot, the experience invites the participant to become the story. Spatial perception (e.g., stairs too steep, rooms too narrow) becomes an expressive narrative layer, embedding affective and epistemic differences into the experience. This is framed within a cyclical narrative informed by Design Thinking [2]: *Empathise* → *Prototype* → *Test* → *Iterate*.

3 Interaction and Implementation

3.1 VR System Overview

The VR system was developed in Unity¹ using the Universal Render Pipeline (URP) to support high-performance rendering on Meta Quest 3.

The experience includes three narrative-driven scenes (see Fig. 1):

- **Heaven:** The player descends as God to interact with the cloud companion “Nebu” to receive and assemble procedurally shaped blocks on the assembly desk. The mortal avatar “Echo” on the desk provides scale reference. Completed buildings are later instantiated in the Mortal World.
- **Mortal World:** By entering a Portal, the player inhabits “Echo” and explores the building from a mortal perspective, reflecting on design impact. The player can return to Heaven through the Portal with user feedback.
- **Echoes Origin:** This starlit space visualises user voices as glowing gems, each containing an audio message and four

AI-generated images² in varied art styles. The space emphasises that creators should value users' needs. In addition, a paintbrush is provided to paint inspiration under the stars.

3.2 Multimodal Interaction

The system involves multimodal interaction, enhancing the affective immersion and sensory experience. Grabable objects are highlighted via ray-based visual cues, supporting intuitive interaction. “Echo” provides a visual reference to gauge the relative scale of building blocks, enhancing intuitive understanding of proportion. Gems emit spatial audio and glow uniquely, revealing narrative audio and AI-generated images when held. Ambient sounds vary by scene, for instance, Heaven features ethereal tones while Mortal World uses nature ambience, which helps players internalise role shifts. Gaze interaction with “Nebu” triggers voiced guidance and light feedback.

3.3 AI-driven Companions

Two AI-driven companions serve as narrative agents and emotional bridges between the creator and the user. “Nebu”, a floating guide between realms, offers contextual prompts and block assistance. “Echo”, a curious mortal avatar, expresses user needs and creative visions. Both facilitate reflection on creation from dual perspectives.

Powered by Convai³, their dialogues are dynamically triggered by location and intent, enhancing spatial storytelling and user engagement. By defining their voices, personalities, and backstories, each character is shaped to fulfil their respective narrative and functional roles. For example, when the player approaches the desk, “Nebu” responds with a guiding prompt “Great! You have reached the assembly desk. Tap me, I'll show you magic bricks.”

4 Conclusion and Future Directions

“Echoes of the Creator” presents a conceptual prototype that invokes reflection on co-creation through immersive storytelling, role-shifting, and multimodal interaction. By enabling a single user to alternate between the roles of a god-like creator and a mortal inhabitant, the system metaphorically explores power, empathy, and design responsibility.

As part of the design iteration process, we conducted informal pilot testing with 8 participants (6 Male, 2 Female), which indicated that the system's embodied role-switching and AI interactions encouraged empathy and collaborative reflection.

Future developments will extend to multi-user settings, customizable building blocks, and user-driven feedback loops, where mortals can mark areas for revision and creators can respond. These features aim to support more flexible, reciprocal, and imaginative collaboration in virtual environments.

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¹<https://unity.com/>

²<https://jimeng.jianying.com/>

³<https://convai.com/>

The project also made use of openly available resources. 3D models from the Unity Asset Store included natural elements (*Low Poly Environment Park*⁴, *Simple Low Poly Nature Pack*⁵), gems (*Low Poly Gems*⁶), a skybox (*3 Skyboxes*⁷), and the mortal avatar (*Character Pack: Free Sample*⁸). Additional resources included 3D hand prefabs from Valem Tutorials⁹, ambient and interaction sounds from FreeSound¹⁰ (e.g., *Soothing Ambient Music for Yoga & Meditation*¹¹, *cultus lake dawn chorus.wav*¹², *Heart collect/Healing (Retro)*¹³, and *Double Drop.mp3*¹⁴), Nebu's speech generated with Hume AI¹⁵, and audio messages created with Luvvoice¹⁶. All original scene compositions, including floating islands, portals, building blocks, and the assembly desk, as well as the spatial layout and overall storytelling structure, were created by the author.

References

- [1] Joseph Campbell. 2008. *The hero with a thousand faces*. Vol. 17. New World Library, Novato, California.
- [2] Christoph Meinel, Larry Leifer, and Hasso Plattner. 2011. *Design thinking: Understand-improve-apply*. Springer, Berlin, Germany.
- [3] Chris Milk. 2015. How Virtual Reality Can Create the Ultimate Empathy Machine. TED Talk. https://www.ted.com/talks/chris_milk_how_virtual_reality_can_create_the_ultimate_empathy_machine
- [4] Gareth W Young, Néill O'Dwyer, and Aljosa Smolic. 2022. Exploring virtual reality for quality immersive empathy building experiences. *Behaviour & information technology* 41, 16 (2022), 3415–3431.
- ⁴<https://assetstore.unity.com/packages/3d/environments/low-poly-environment-park-242702>
- ⁵<https://assetstore.unity.com/packages/3d/environments/landscapes/simple-low-poly-nature-pack-157552>
- ⁶<https://assetstore.unity.com/packages/3d/props/low-poly-gems-245515>
- ⁷<https://assetstore.unity.com/packages/2d/textures-materials/sky/3-skyboxes-25142>
- ⁸<https://assetstore.unity.com/packages/3d/characters/humanoids/character-pack-free-sample-79870>
- ⁹<https://www.youtube.com/@ValemTutorials>
- ¹⁰<https://freesound.org/>
- ¹¹<https://freesound.org/people/Matio888/sounds/796500/>
- ¹²<https://freesound.org/people/tim.kahn/sounds/200256/>
- ¹³<https://freesound.org/people/ReincarnatedEchoes/sounds/644306/>
- ¹⁴<https://freesound.org/people/Cyclomatic/sounds/419551/>
- ¹⁵<https://platform.hume.ai/>
- ¹⁶<https://luvvoice.com/>